Report on : Plasma protein profiling predicts cancer in patients with non-specific symptoms

Melika Janmohammadifiroz, Department of Biology, College of Fundamental Science, Azad University, Qaemshahr Branch, Qaemshahr, Iran; E-Mail: melika.janmohammadifiroz@iau.ir

Abstract

Discrimination between malignancy and non-malignant conditions in patients poses a significant challenge among general practitioners (GPs), especially in patients with non-specific symptoms. This study evaluated the plasma protein profiling technique using proximity extension assay (PEA) technology to predict cancer risk. 1,463 plasma proteins were assessed in the discovery cohort with 456 participants and the Replication cohort with 238 participants. Subsequently, 29 protein signatures were identified as associated with new cancer diagnoses. The model developed in the study demonstrated strong performance, particularly in Area Under the Curve (AUC), achieving 0.80 in the discovery and 0.82 in the replication cohorts. Additionally, cancer is effectively distinguished from autoimmune, inflammatory, and infectious diseases. These findings demonstrate that plasma protein profiling would be an effective triage tool that would enhances prioritizing high-risk patients for advanced imaging and minimizing unnecessary exposure to invasive diagnostic approaches.

Keywords: Cancer Biomarkers; Machine Learning Classification; Non-specific Symptoms; Plasma Proteomics; Proximity Extension Assay (PEA).

Introduction

Non-specific symptoms, described by general malaise, extreme fatigue, reduced appetite, and pathological laboratory findings, are a complex issue because they are not immediately and directly indicative of organ-specific cancer. This vague condition makes early diagnosis challenging, which is crucial in cancer treatment. To address this issue, an approach using the blood protein plasma profile is introduced to predict the probability of cancer risk for patients with nonspecific symptoms. This is a fast-tracking triage technique, assisting general practitioners (GPs) in determining which patient group requires invasive imaging and other diagnostic approaches. One strength of this study is that the model distinguishes whether these symptoms result from cancer or other conditions such as infection, inflammation, autoimmune disease, or non-cancerous conditions. It has been highlighted by previous studies that plasma protein panels could enhance the diagnosis of cancer, while the specificity and sensitivity of individual protein markers and false positives require further investigation. (1) (2) (3) (4) In addition to this, the necessity of validating plasma proteins and the caution against relying only on this approach have been emphasized. (1) Despite the above-mentioned limitations, plasma protein profiling for cancer diagnosis has represented promising results. (1) (3) In the Wannberg study, (5) The strength of the technique in diagnosing 160 malignancies and 296 non-malignant conditions is also effectively demonstrated. (5) In this report, the methodology of this technique will be described.

Study population

The study population comprised two independent cohorts of patients referred to fast-track diagnostic pathways for non-specific symptoms. The discovery phase analyzed the plasma proteome of 456 patients from the MEDECA cohort in Stockholm, Sweden. Females comprised 55% of this cohort, with a median age of 71 years (IQR 60–78). Non-specific symptoms in this cohort included general malaise, extreme fatigue, weight loss, pathological laboratory values or radiological findings suggestive of metastasis without a primary tumor, and other criteria. After a six-month diagnostic work-up, 160 patients in the discovery cohort were diagnosed with cancer, with hematologic malignancies being the most common cancer type; all others were diagnosed with non-malignant conditions. These findings were validated in an independent replication cohort, ALLVOS in Örebro, Sweden, consisting of 238 patients, of whom 50% were female, with a median age of 72 years. The inclusion criteria were similar, except for the exclusion of patients with radiological signs of malignancy. In this replication group, 35 patients were diagnosed with cancer.

Protein profiling

The Olink Proximity Extension Assay (PEA) is a high-throughput, antibody-based proteomics technology. In PEA, oligonucleotide-labeled antibodies bind to their target antigens. After hybridization, the associated DNA sequences between them will be amplified and quantified using NGS by the Illumina NovaSeq 6000 platform. This technique provides simultaneous quantification of multiple proteins with high sensitivity and specificity for biomarker discovery and disease profiling, particularly in neurology, oncology, and immunology. (6) (7) (8) (9) The output of this method is NPX (Normalized Protein expression), which is normalized to the corresponding Extension Control signal and subsequently log2-transformed, providing a relative Quantification. Each one-unit increase in NPX represents a doubling of protein concentration. To remove systematic intensity differences and enable comparability across the dataset, the median NPX of each assay on each plate was centered at zero.

Statistical analysis and machine learning

Exploratory Data Analysis

To demonstrate the data pattern and structure, the data was projected into simpler principal components. The majority of variation in the dataset is captured in PC1 (about 48% to 65%, respectively, in replication and discovery) and PC2. In addition to this, Distinct clustering patterns were illustrated, including patients later diagnosed with cancer, non-cancerous conditions such as autoimmune, infectious, and inflammatory diagnoses, and those who received no diagnosis. Accordingly, a clear left-right separation between groups along PC1 was generated that discriminate no diagnosis cases and other non-malignant diagnoses, and the cancer diagnoses group. In both cohort the loading scores for PC1 for the proteins were similar, and NPX Levels were higher in cancer compared to other groups.

Differential Expression Analysis

Differential expression analysis was performed between patient groups and controls by comparing their protein abundance among them using the NPX value and the linear models limma. In order to define the differential protein expression between the cancer and non-cancer group, the model limma was used, which was calculated as log 2-fold change (logFC). That shows more stable variance, and enhances the comparison of protein expression level. In this model, some filters, such as age, sex, and p-value using Benjamini–Hochberg (FDR), were applied. In addition to this log fold change (logFC)>0.5 filter was established. Each one-unit increase in NPX results in greater than 1 in the Odds Ratio value (OR), which indicates the cancer risk probability, and this is the model output. Several of the identified proteins were characterized as cancer-related proteins, according to the Human Protein Atlas (v24.proteinatlas.org).

Model design

A multivariate pan-cancer classification model was developed using penalized logistic regression (LASSO) based on proteomics data .The protein data were scaled between 0 and 1 based on their importance. The Regularization was applied to decrease overfit risk with feature selection and eliminate less informative features.

The data was randomly partitioned by allocating 70% of the data to the training set and 30 % of the data to the test set. Three initial filtering criteria to reduce the number of features before model fitting were applied to a protein needed to meet at least one of the following: (i) a mean NPX difference greater than 1, (ii) an adjusted p-value < 1E−6 from a t-test, or (iii) an area under the receiver operating curve (AUC) > 0.7 for individual proteins. To optimize the model, 10-fold cross-validation was performed on the training data by dividing the data into ten distinct subsets. In each iteration, the model was trained on nine folds, and the tenth fold was allocated for validation. Through these selections, 29 proteins were identified as input for the model.

Model performance

The model demonstrated robust performance with an Area Under the Curve (AUC) of 0.80 in the discovery cohort and maintained performance with an AUC of 0.82 when applied to the replication cohort. In the subgroup analysis, the AUC score for distinguishing cancer from the No Diagnosis group was 0.86, while the scores for Infectious, Autoimmune, and Inflammatory diseases were 0.76, 0.75, and 0.67, respectively. Unlike other results, the discrimination between Inflammatory and Cancer conditions was not robust enough. Among the identified proteins, six proteins were selected as the most important proteins, and all of these proteins were identified as upregulated proteins in cancer patients. Biologically, their role such as proliferation, DNA repair, and cell adhesion, are established. AGR2, KRT-19, CEACAM5 (CEA), RRM2, PARP1, PSIP1.

Strengths and Limitations of the Study

This study had two primary strengths; firstly, the control group was not healthy; instead, the study included patients with non-malignant conditions, such as infection, inflammation, autoimmune disease, or non-cancerous conditions, which more efficiently represent a real-world clinical setting. Secondly, samples were collected before the cancer diagnostic work-up. Despite these advantages, there are several limitations in this study. There are many cases in the study with later diagnosed metastatic which hinder generalizability in cohorts. Furthermore, the replication cohort contained a lower proportion of patients later diagnosed with cancer, as it excluded patients with radiological findings of metastases without an identifiable primary tumor. However, the model's ability to perform with equal performance in the replication cohort despite this difference is considered a significant advantage.

Future Research Recommendations

Some recommendations for future studies have been highlighted. Firstly, focusing on populations with lower cancer prevalence is significant in terms of generalizability and clinical utility. Secondly, using absolute protein concentrations for clinical implementation is recommended. Finally, validation of the model in a healthy population and direct comparison with other diagnostic blood test-based platforms, including commercial pan-cancer tests such as GRAIL's Galleri and Cancerguard.

Conclusion

This study demonstrated that plasma protein profiling, applying machine learning, is an effective approach to predict cancer in patients with vague symptoms. The model efficiently differentiates protein expression patterns between groups with cancer and non-cancer conditions before clinical diagnosis, with strong performance in both cohorts. Although more clinical validation studies are required, this method is a minimally invasive, strong triage tool that could recognize and prioritize high-risk patients significantly and refer them for rapid and advanced medical investigation.

References

1. Videmark AN, Christensen IJ, Feltoft CL, Villadsen M, Borg FH, Jørgensen BM, et al. Combined plasma C-reactive protein, interleukin 6 and YKL -40 for the detection of cancer and prognosis in patients with serious nonspecific symptoms and signs of cancer. Cancer Med [Internet]. 2023 Mar [cited 2026 Feb 12];12(6):6675–88. Available from: https://onlinelibrary.wiley.com/doi/10.1002/cam4.5455

2. Giai Gianetto Q, Michel V, Douché T, Nozeret K, Zaanan A, Colussi O, et al. Plasma Protein Biomarkers to Detect Early Gastric Preneoplasia and Cancer: A Prospective Study. Int J Mol Sci [Internet]. 2025 Oct 17 [cited 2026 Feb 12];26(20):10114. Available from: https://www.mdpi.com/1422-0067/26/20/10114

3. Álvez MB, Edfors F, Von Feilitzen K, Zwahlen M, Mardinoglu A, Edqvist PH, et al. Next generation pan-cancer blood proteome profiling using proximity extension assay. Nat Commun [Internet]. 2023 Jul 18 [cited 2026 Feb 12];14(1):4308. Available from: https://www.nature.com/articles/s41467-023-39765-y

4. Landegren U, Hammond M. Cancer diagnostics based on plasma protein biomarkers: hard times but great expectations. Mol Oncol [Internet]. 2021 Jun [cited 2026 Feb 13];15(6):1715–26. Available from: https://febs.onlinelibrary.wiley.com/doi/10.1002/1878-0261.12809

5. Wannberg F, Álvez MB, Qvick A, Pongracz T, Aguilera K, Adolfsson E, et al. Plasma protein profiling predicts cancer in patients with non-specific symptoms. Nat Commun [Internet]. 2025 Dec 29 [cited 2026 Feb 4];17(1):151. Available from: https://www.nature.com/articles/s41467-025-67688-3

6. Yang L, Cao M, Tian J, Cui P, Ai L, Li X, et al. Identification of Plasma Inflammatory Markers of Adolescent Depression Using the Olink Proteomics Platform. J Inflamm Res [Internet]. 2023 Oct [cited 2026 Feb 15];Volume 16:4489–501. Available from: https://www.dovepress.com/identification-of-plasma-inflammatory-markers-of-adolescent-depression-peer-reviewed-fulltext-article-JIR

7. Shraim R, Diorio C, Canna SW, Macdonald-Dunlop E, Bassiri H, Martinez Z, et al. A Method for Comparing Proteins Measured in Serum and Plasma by Olink Proximity Extension Assay. Mol Cell Proteomics [Internet]. 2025 Jul [cited 2026 Feb 15];24(7):101000. Available from: https://linkinghub.elsevier.com/retrieve/pii/S1535947625000994

8. Carlyle BC, Kitchen RR, Mattingly Z, Celia AM, Trombetta BA, Das S, et al. Technical Performance Evaluation of Olink Proximity Extension Assay for Blood-Based Biomarker Discovery in Longitudinal Studies of Alzheimer's Disease. Front Neurol [Internet]. 2022 Jun 6 [cited 2026 Feb 15];13:889647. Available from: https://www.frontiersin.org/articles/10.3389/fneur.2022.889647/full

9. Christensen TD, Maag E, Madsen K, Lindgaard SC, Nielsen DL, Johansen JS. Determination of temporal reproducibility and variability of cancer biomarkers in serum and EDTA plasma samples using a proximity extension assay. Clin Proteomics [Internet]. 2022 Dec [cited 2026 Feb 15];19(1):39. Available from: https://clinicalproteomicsjournal.biomedcentral.com/articles/10.1186/s12014-022-09380-y